

Python — for Loops & range()

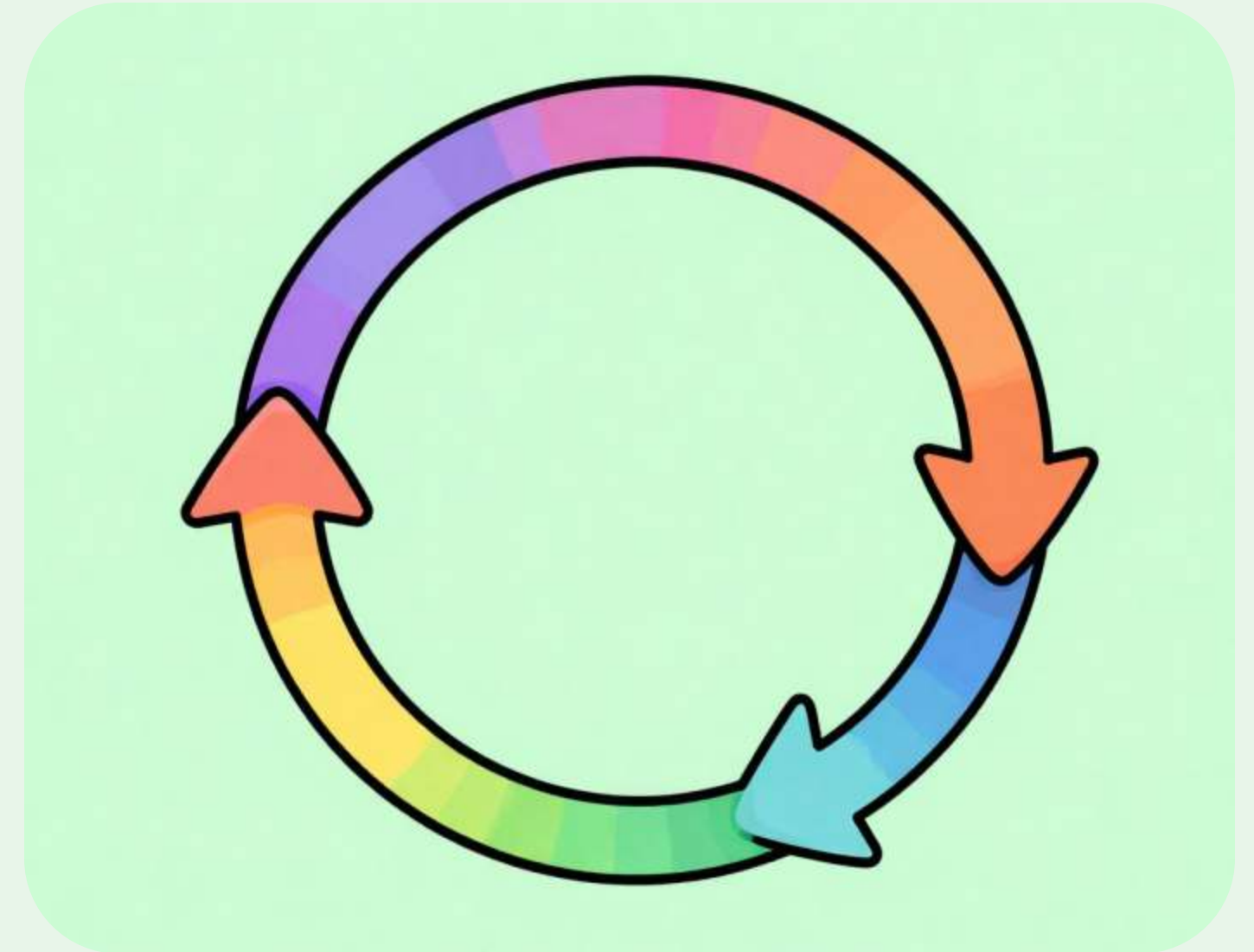
Learn how Python repeats actions using for loops and the range() function!



What is a Loop?

A loop repeats a block of code multiple times.

Instead of writing the same line again and again, we use a loop!



Why Do We Need Loops?

Imagine writing 'Hello' 100 times!

A loop does it in just 2 lines of code.
That's the power of loops!



The for Loop

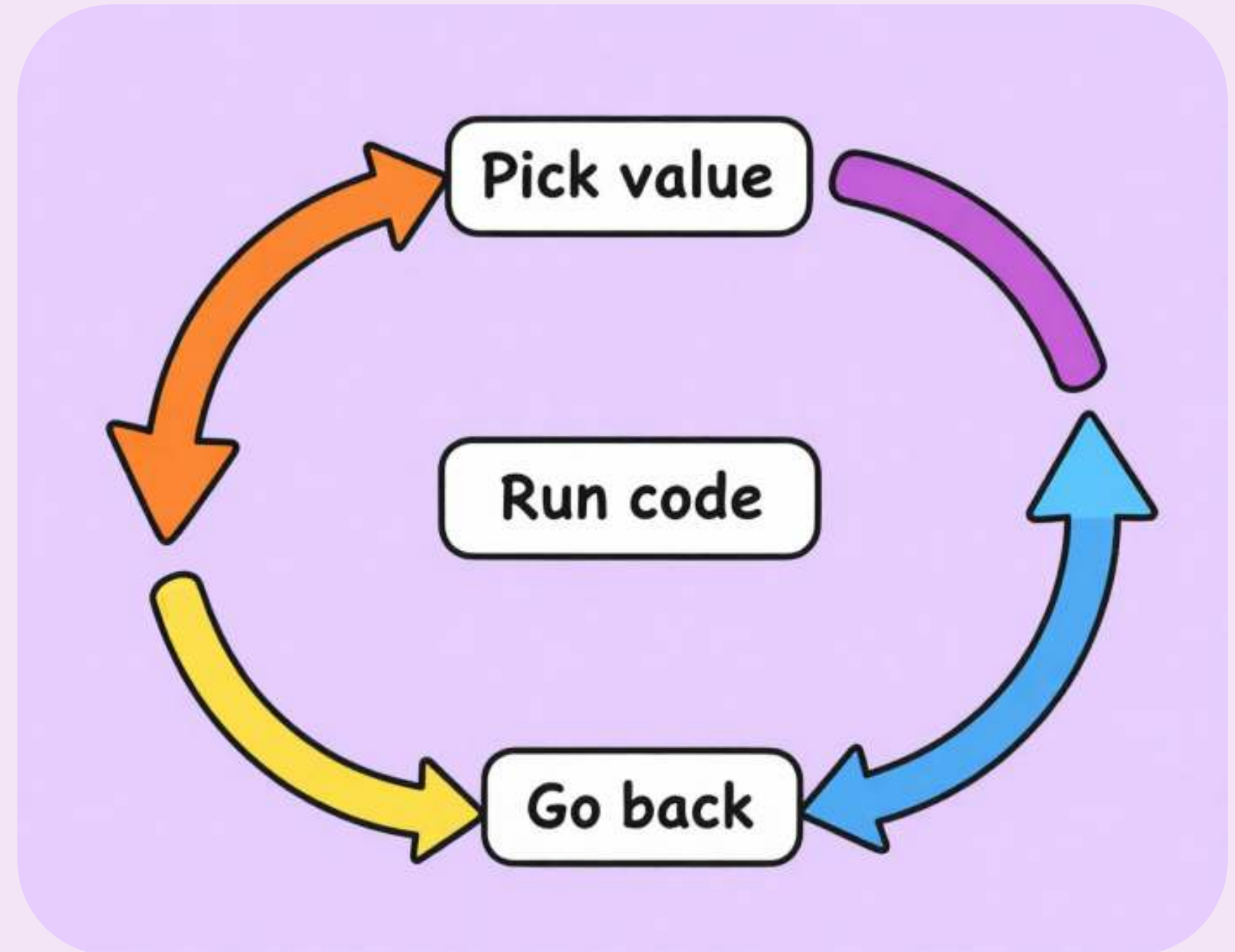
```
for i in range(5):  
    print(i)
```

This prints: 0, 1, 2, 3, 4



How for Loop Works

1. Python picks the first value
2. Runs the indented code
3. Goes back for the next value
4. Repeats until done!



Let's Try: Print 1 to 5

```
for num in range(1, 6):  
    print(num)
```

Output: 1 2 3 4 5



What is range()?

range() generates a sequence of numbers.

It tells the loop how many times to repeat.



range() with One Number

`range(5)` gives: 0, 1, 2, 3, 4

It starts at 0 and stops before 5.
5 numbers total!



range() with Two Numbers

`range(2, 7)` gives: 2, 3, 4, 5, 6

First number = start

Second number = stop (not included)



range() with Three Numbers (Step)

`range(0, 10, 2)` gives: 0, 2, 4, 6, 8

The third number is the step.
It skips numbers!



Loop Through a List

```
fruits = ['mango', 'banana', 'apple']  
for f in fruits:  
    print(f)
```

It loops through each item!



Fun Pattern: Stars!

```
for i in range(1, 6):  
    print('*' * i)
```

Output:

*

**



Counting Backwards

```
for i in range(5, 0, -1):  
    print(i)
```

Output: 5 4 3 2 1
Countdown!



Common Mistakes

Common mistakes to avoid:

1. Forgetting the colon :
2. Not indenting the loop body
3. Wrong range numbers



Challenge Time!

Try these challenges:

1. Print your name 5 times
2. Print even numbers 2 to 20
3. Make a triangle with #



Great Job! Keep Coding!

You learned for loops and range()!

Now you can make Python
repeat anything.

Keep practising and have fun!

